Computer Networks (CL3001)

CN LAB 05

Submitted by:

Syeda Sara Ali

23K-0070

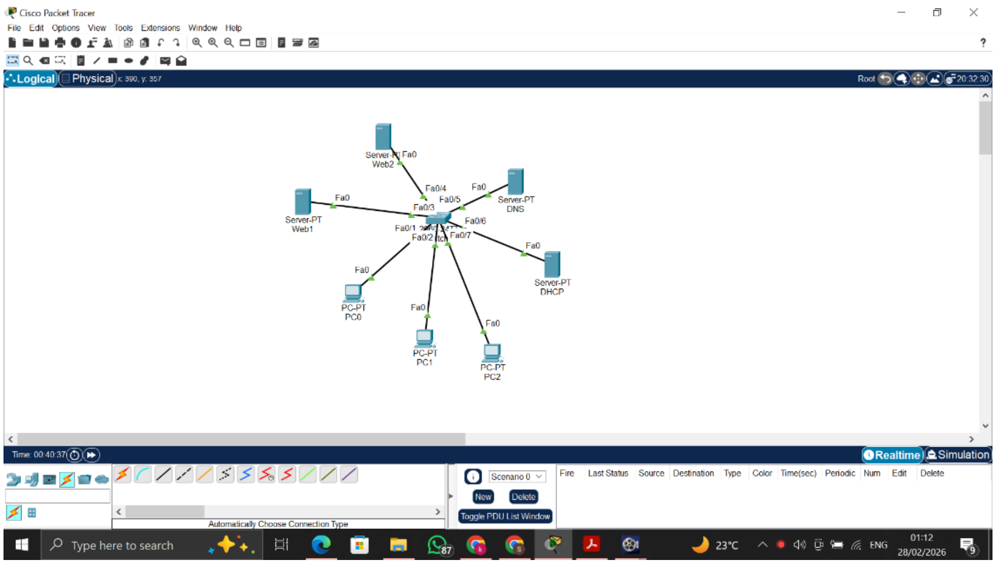
BS-AI (6A)

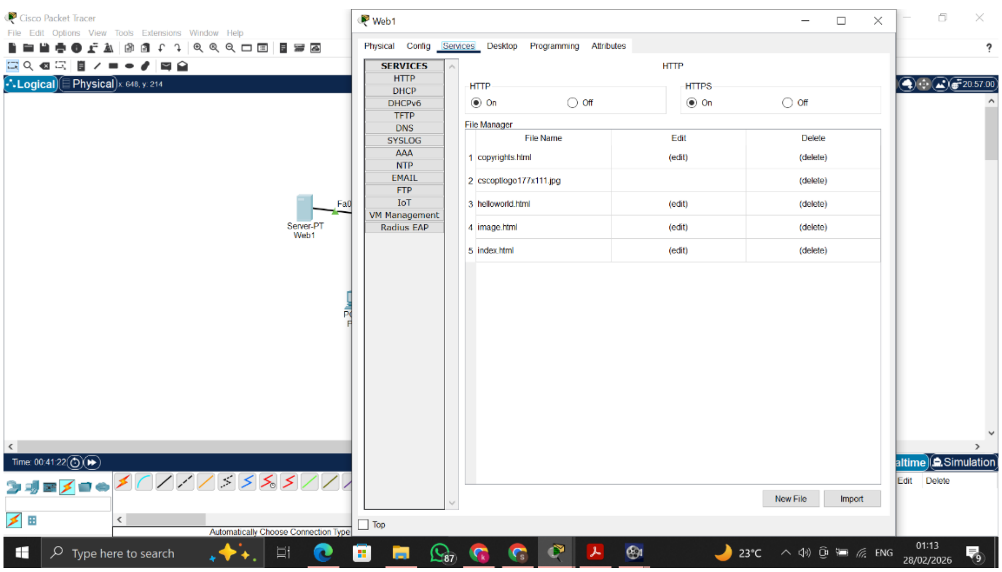
Submitted to:

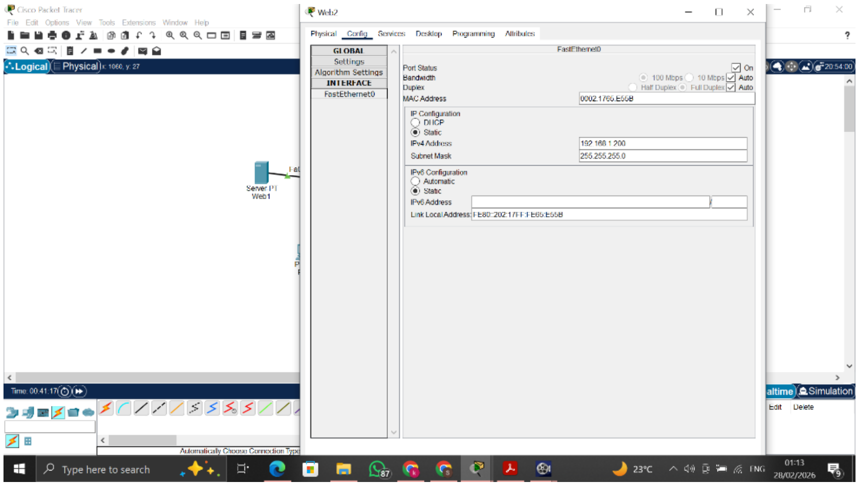
Sir Sameer Faisal

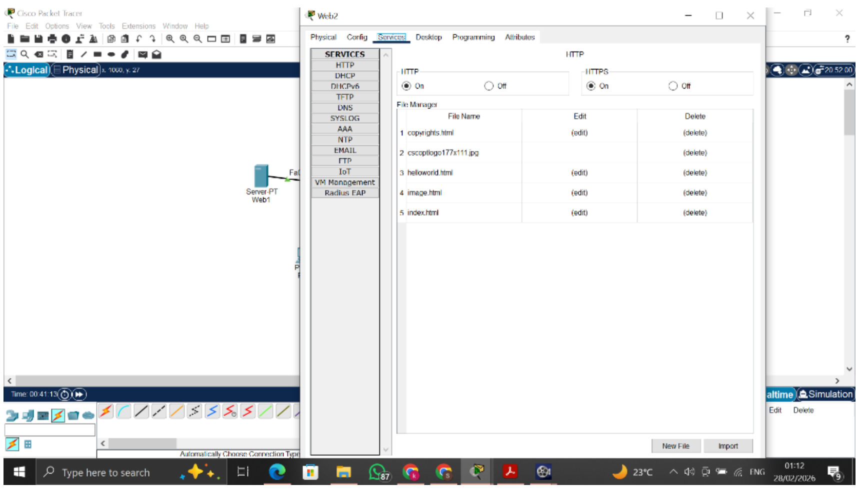
Lab Tasks 1:

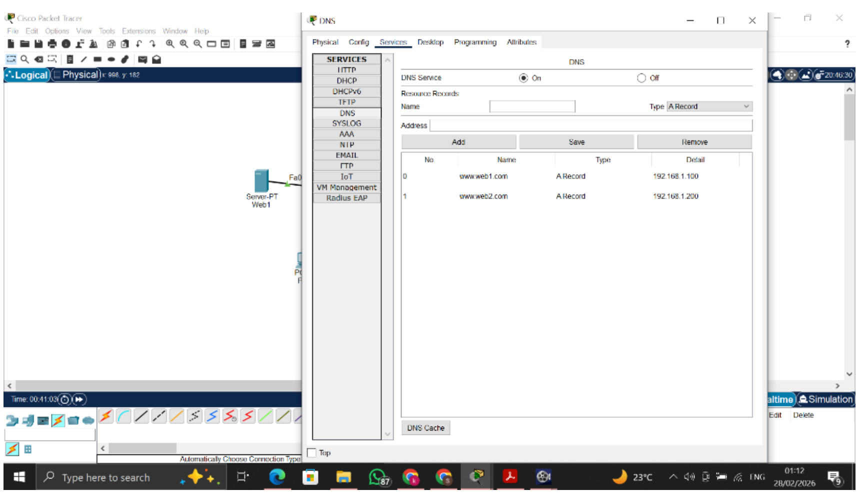
Create the following network with DHCP server. Send DNS packets in your network. Add some more servers in your network.

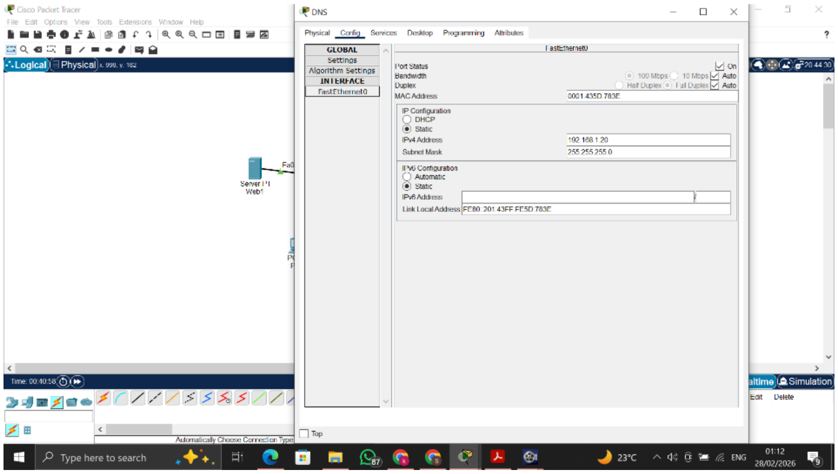


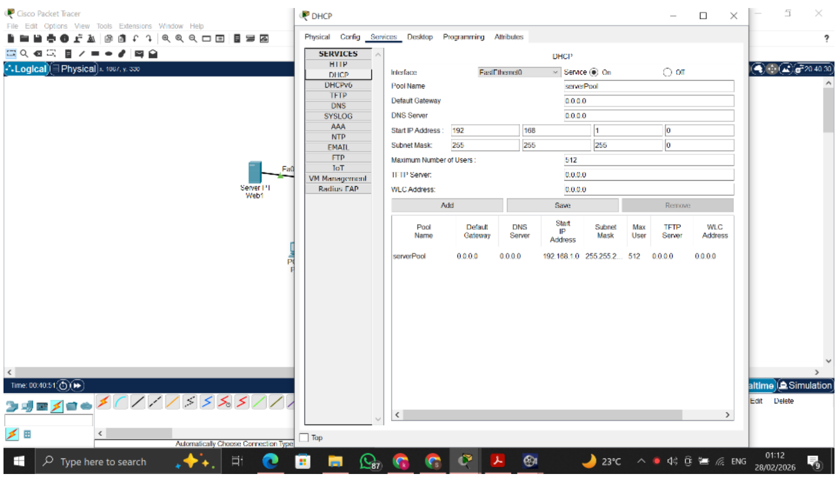


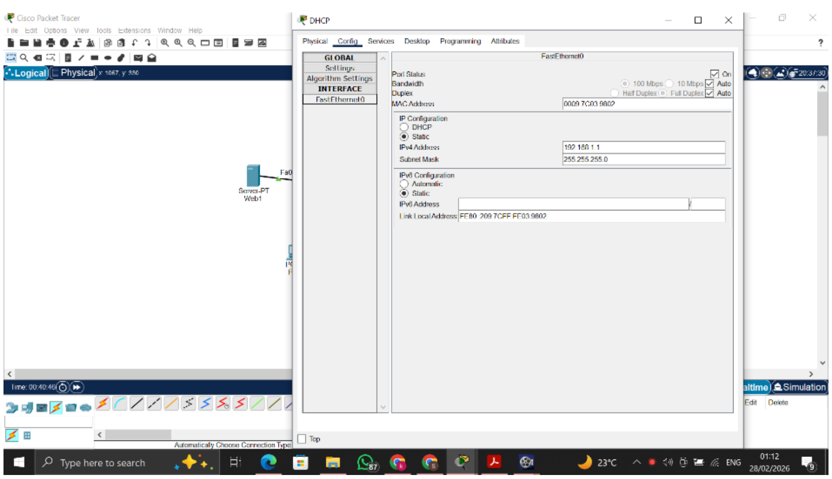

Web2
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.200
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Type here to search

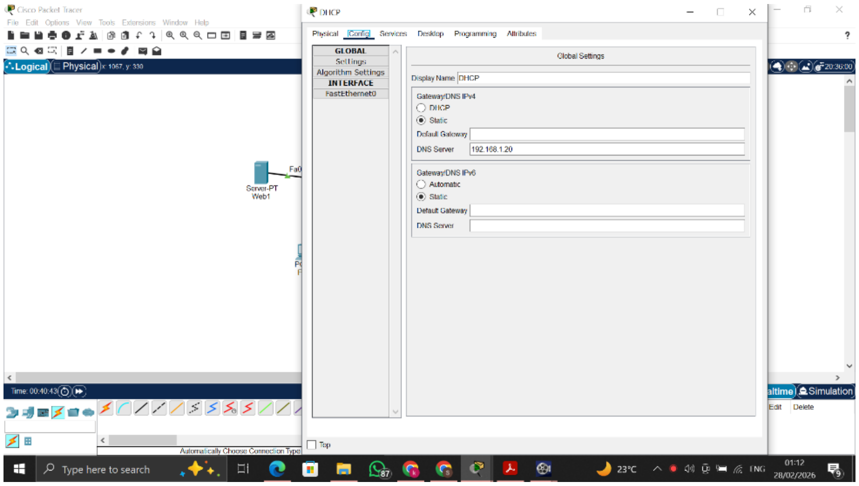

Web2
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html
2 cscoptlogo177x111.jpg
3 helloworld.html
4 image.html
5 index.html
New File
Import
Type here to search

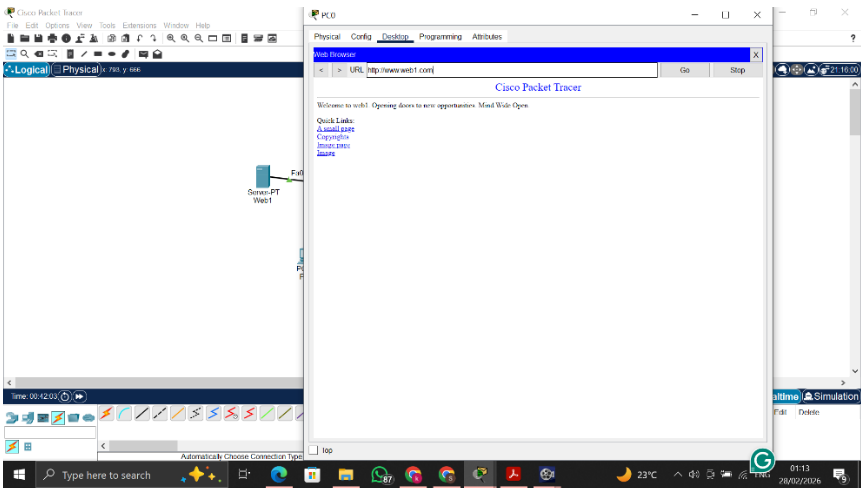

DNS
SERVICES
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
www.web1.com
A Record
192.168.1.100
1
www.web2.com
A Record
192.168.1.200
DNS Cache
Type here to search

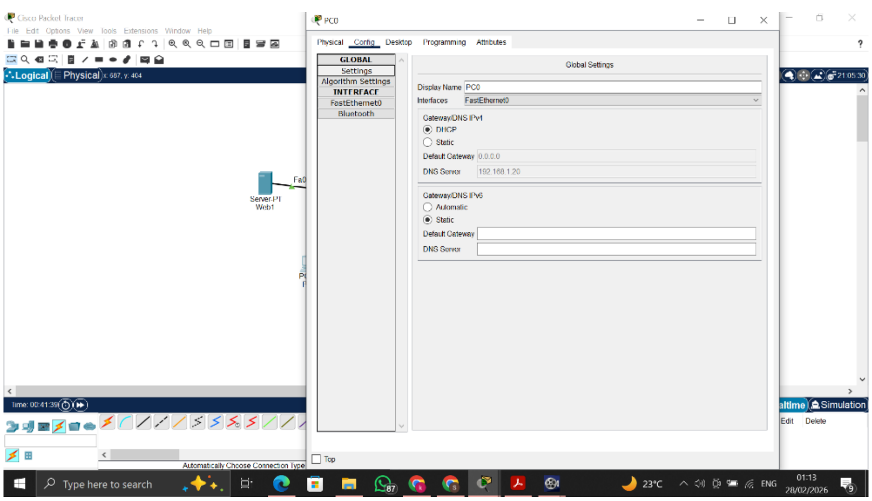

DNS
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.20
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Top
Type here to search
23°C
ENG
01:12
28/02/2026

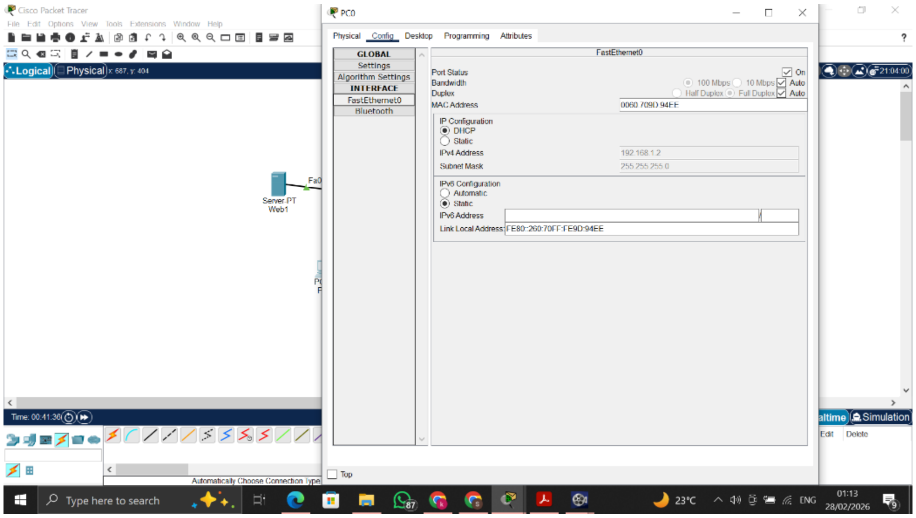

DHCP
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
192
168
1
0
Subnet Mask:
255
255
255
0
Maximum Number of Users :
512
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
192.168.1.0
255.255.2...
512
0.0.0.0
0.0.0.0
Top
Type here to search
23°C
ENG
01:12
28/02/2026

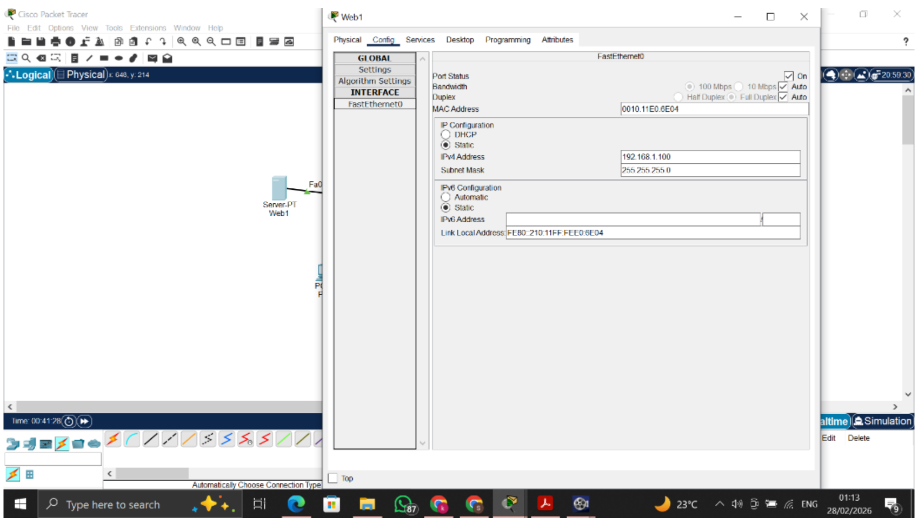

DHCP
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Server-PT
Web1
Top
Type here to search
23°C
ENG
01:12
28/02/2026


DHCP
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name DHCP
Gateway/DNS IPv4
DHCP
Static
Default Gateway
DNS Server 192.168.1.20
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Server-PT
Web1
Simulation
Type here to search
23°C
ENG
01:12
28/02/2026


PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL http://www.web1.com
Go
Stop
Cisco Packet Tracer
Quick Links:
Server-PT
Web1
Simulation
Type here to search
23°C
01:13
28/02/2026

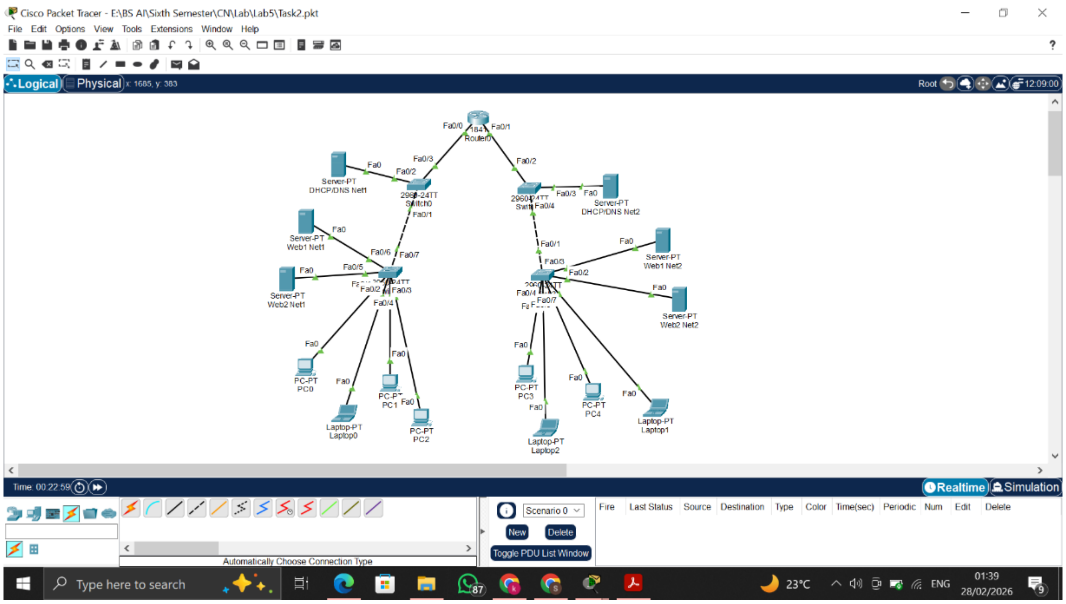

PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name PC0
Interfaces FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway 0.0.0.0
DNS Server 192.168.1.20
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Simulation
Type here to search
23°C
ENG
01:13
28/02/2026

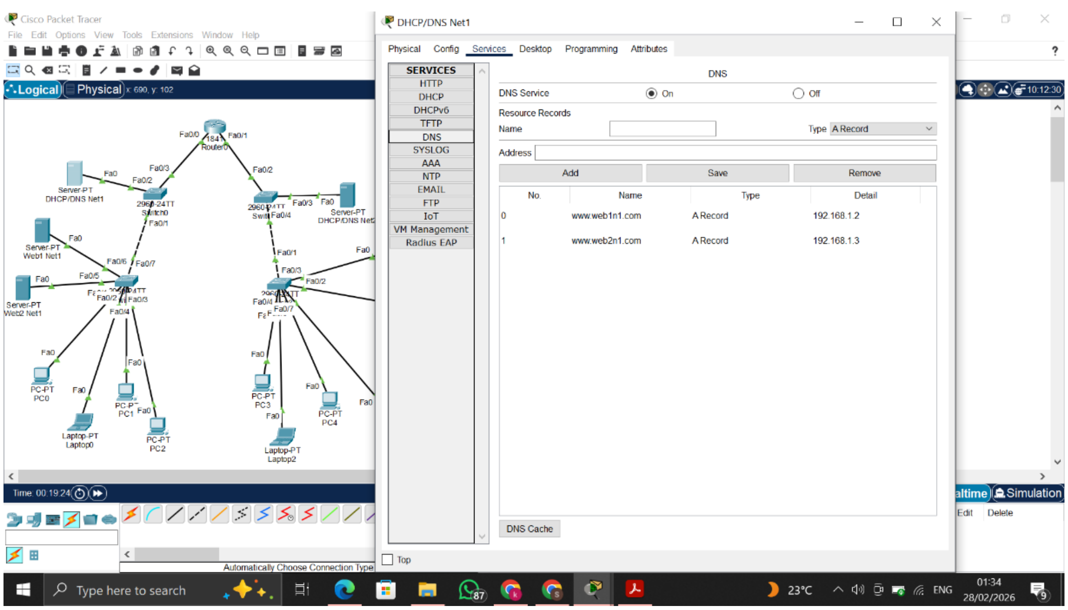

Cisco Packet Tracer
PC0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0060.709D.94EE
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::260:70FF:FE9D:94EE
Top
Server-PT
Web1
Simulation
Type here to search
23°C
ENG
01:13
28/02/2026

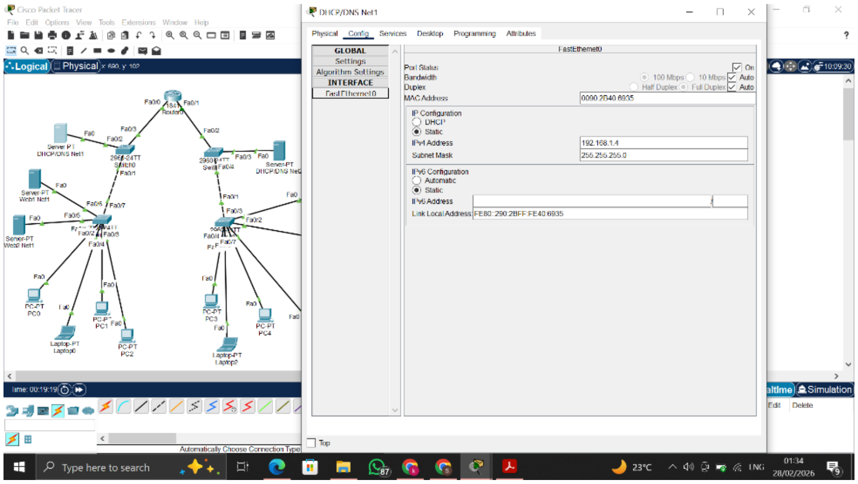

Cisco Packet Tracer
Web1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0010.11E0.6E04
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.100
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::210:11FF:FEE0:6E04
Top
Server-PT
Web1
Simulation
Type here to search
23°C
ENG
01:13
28/02/2026

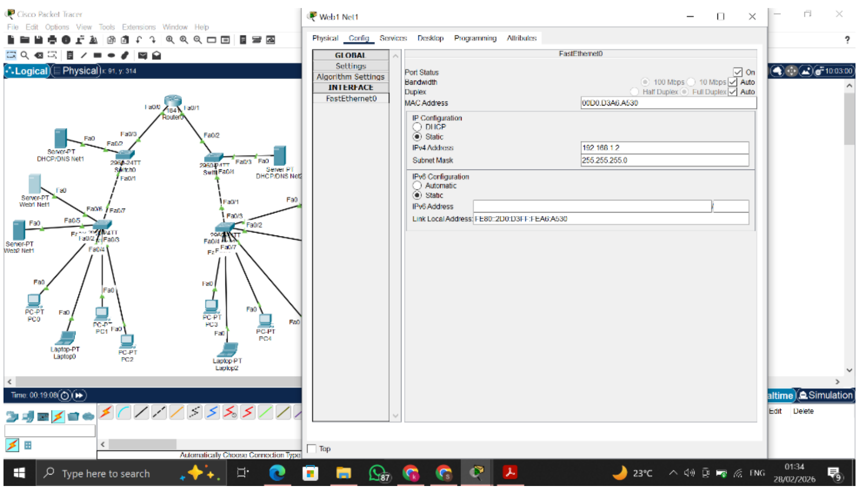
Lab Tasks 2:

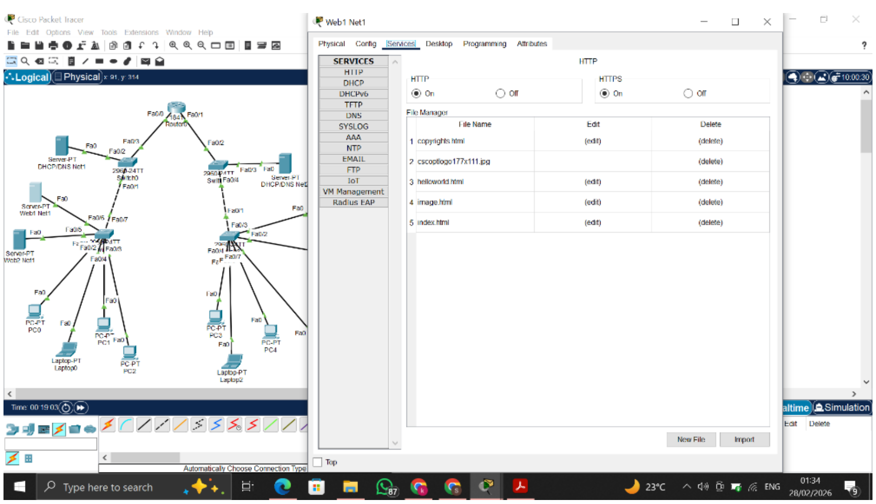
Implement the given topology. Add some more web servers in your network. Implement DNS & add records of your web servers. Use DHCP for IP assignment.

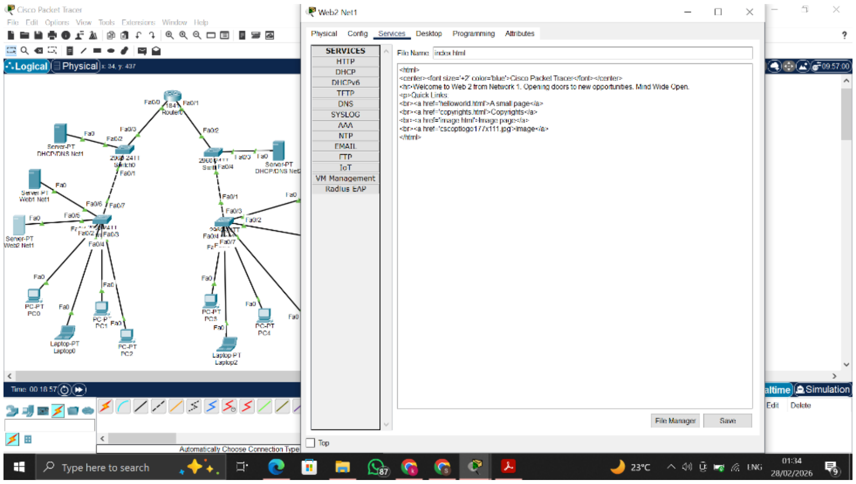





DHCP/DNS Net1
Physical
Config
Services
Desktop
Programming
Attributes
FastEthernet0
MAC Address 0090.2B40.6935
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.4
Subnet Mask 255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::290:2BFF:FE40:6935
Type here to search
01:34
28/02/2026


Web1 Net1
Physical
Config
Services
Desktop
Programming
Attributes
FastEthernet0
MAC Address 00D0.D3A6.A530
IP Configuration
DHCP
Static
IPv4 Address 192.168.1.2
Subnet Mask 255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address FE80::2D0:D3FF:FEA6:A530
Type here to search
01:34
28/02/2026


Web1 Net1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html
2 cscoptlogo177x111.jpg
3 helloworld.html
4 image.html
5 index.html
New File
Import
Type here to search
01:34
28/02/2026

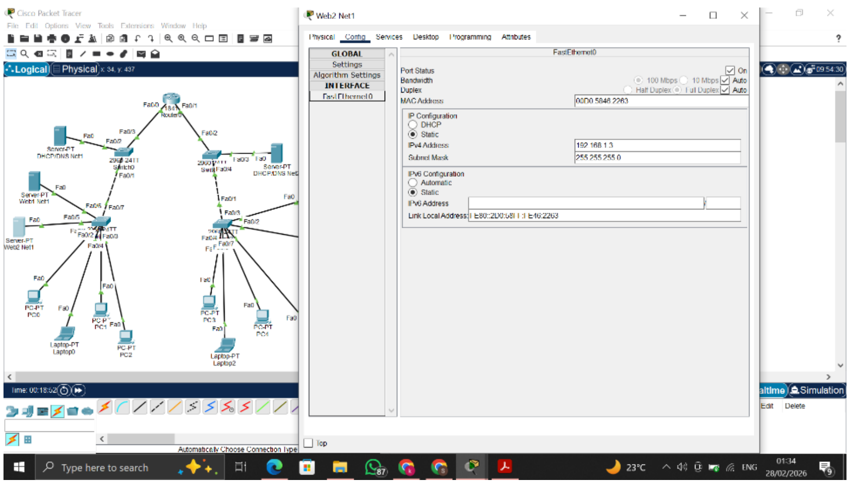

Web2 Net1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
File Name: index.html
<html>
<center>Cisco Packet Tracer</center>
<hr>Welcome to Web 2 from Network 1. Opening doors to new opportunities. Mind Wide Open.
<p>Quick Links:

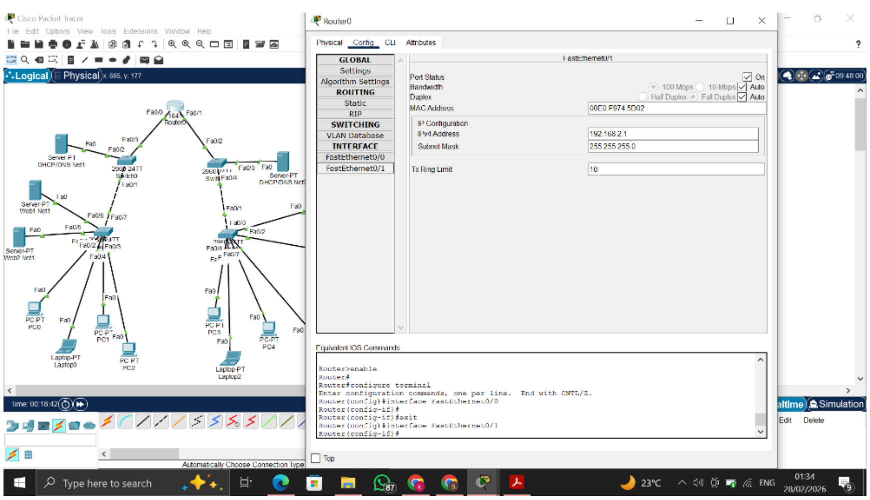
A small page

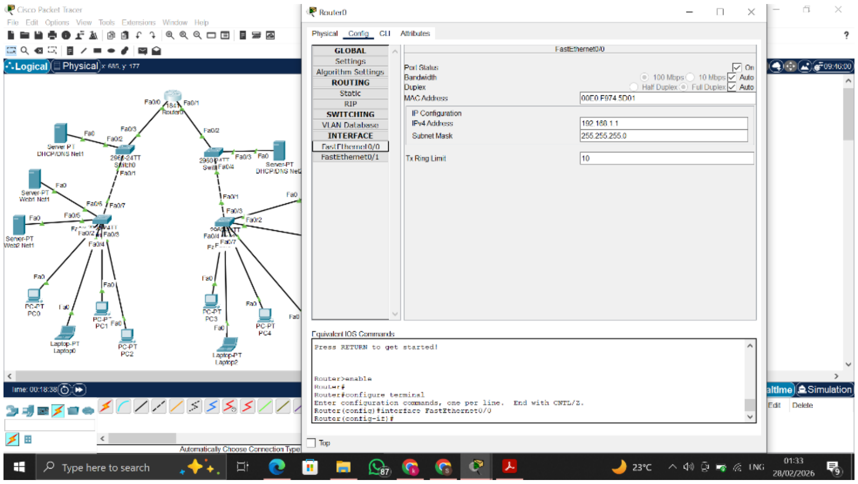
Copyrights

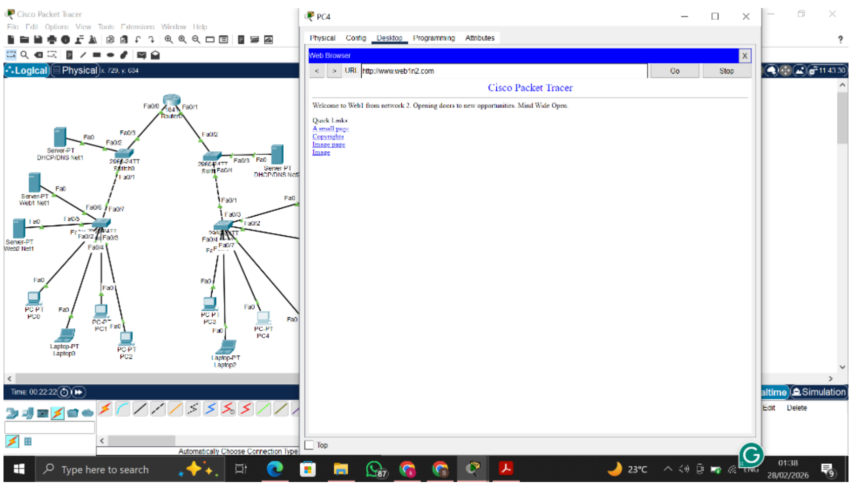
Image page

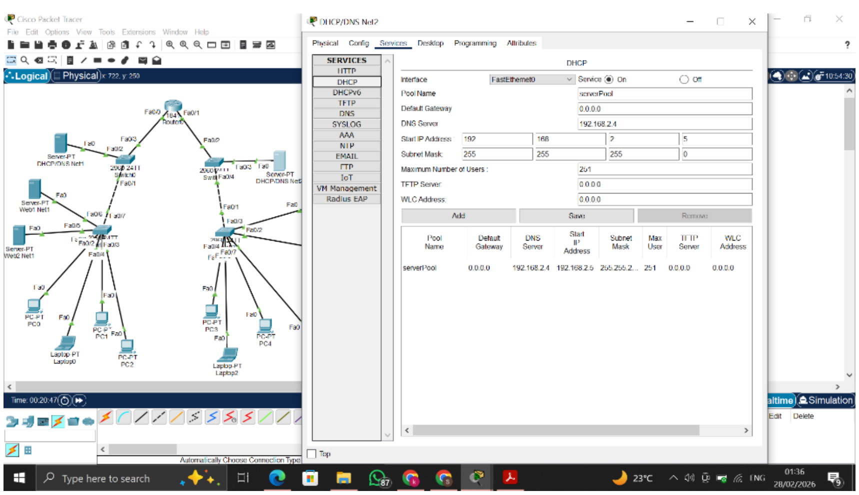
Image
</html>
File Manager
Save
Top
Type here to search
23°C
ENG
01:34
28/02/2026

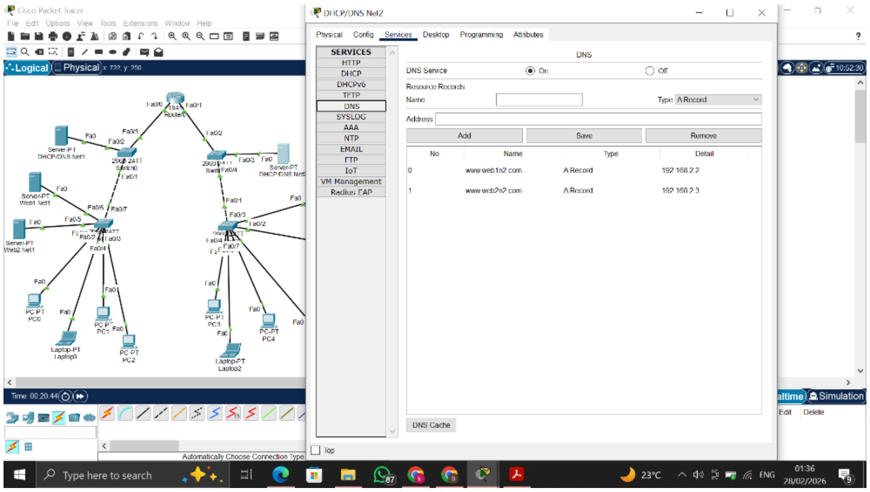

Web2 Net1
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.5846.2263
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2D0:58FF:FE46:2263
Top
Type here to search
23°C
ENG
01:34
28/02/2026

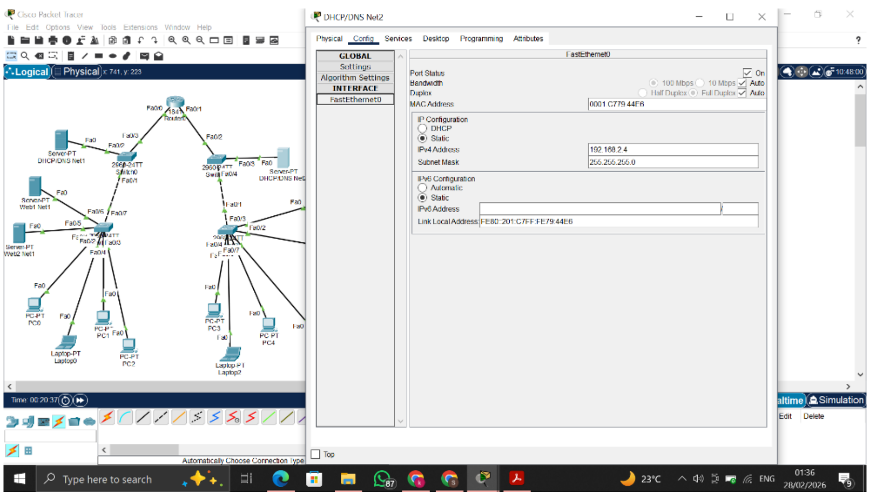

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
FastEthernet0/1
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F974.5D02
IP Configuration
IPv4 Address
192.168.2.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10
Equivalent IOS Commands
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#
Top
Type here to search
23°C
ENG
01:34
28/02/2026

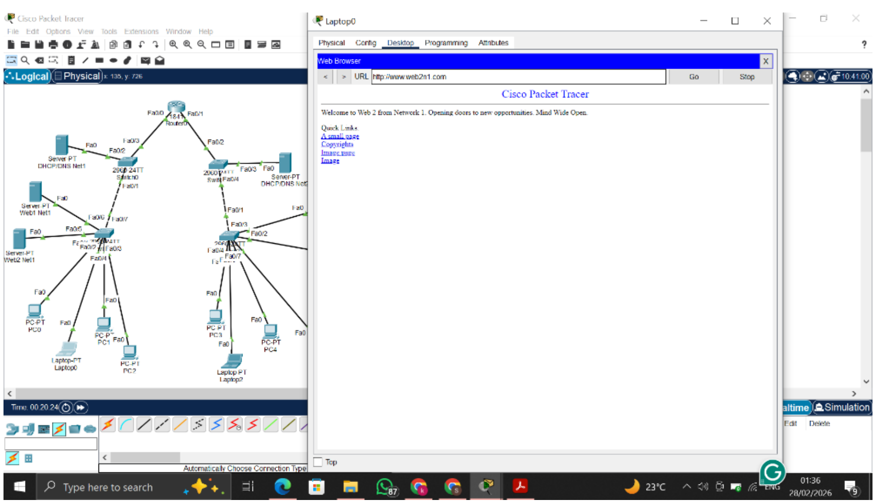

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Port Status
Bandwidth
Duplex
MAC Address
IP Configuration
IPv4 Address
Subnet Mask
Tx Ring Limit
192.168.1.1
255.255.255.0
10
Equivalent IOS Commands
Press RETURN to get started!
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#
Type here to search
23°C
01:33
28/02/2026

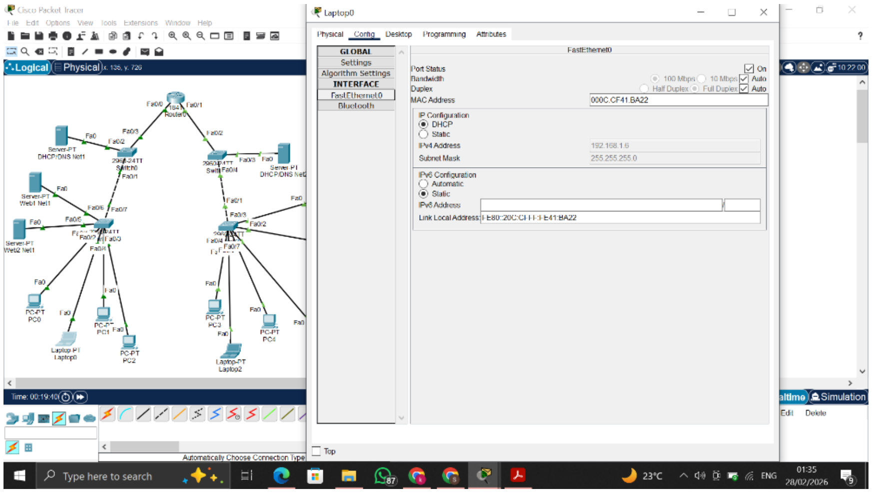

PC4
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL
http://www.web1n2.com
Go
Stop
Cisco Packet Tracer
Welcome to Web1 from network 2. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image
Type here to search
23°C
01:38
28/02/2026

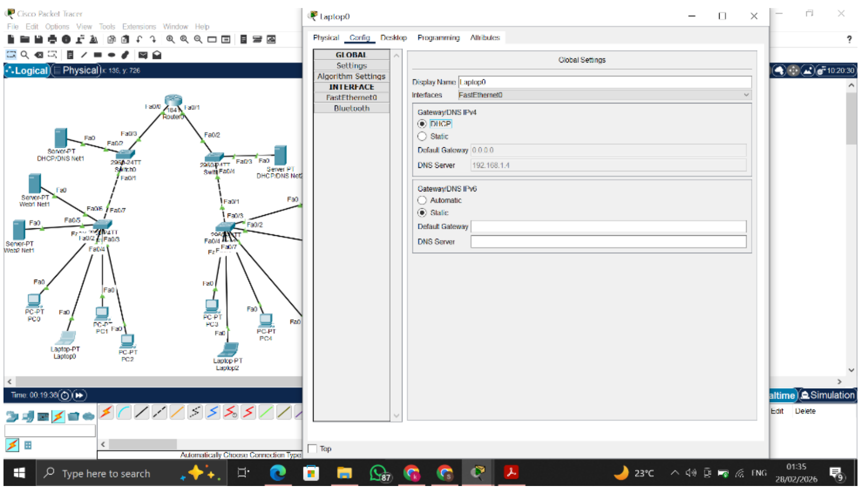

DHCP/DNS Net2
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
192.168.2.4
Start IP Address
192
168
2
5
Subnet Mask
255
255
255
0
Maximum Number of Users :
251
TFTP Server
0.0.0.0
WLC Address
0.0.0.0
Add
Save
Remove
serverPool
0.0.0.0
192.168.2.4
192.168.2.5
251
Type here to search
23°C
01:36
28/02/2026

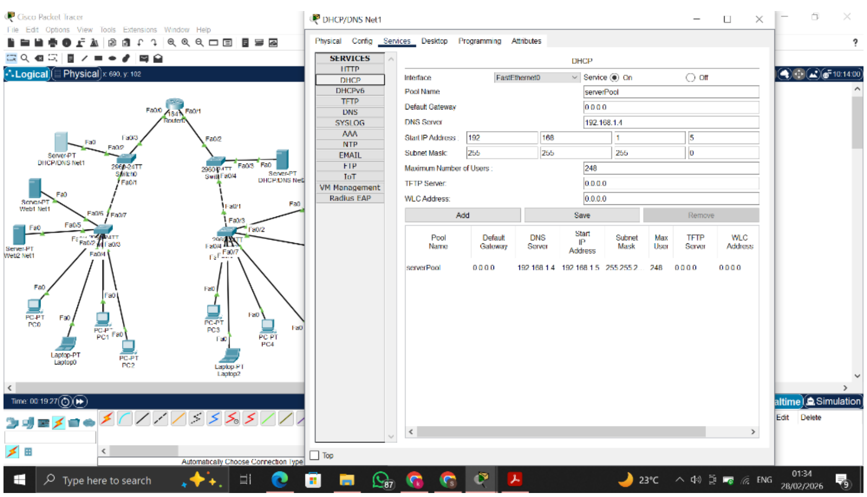






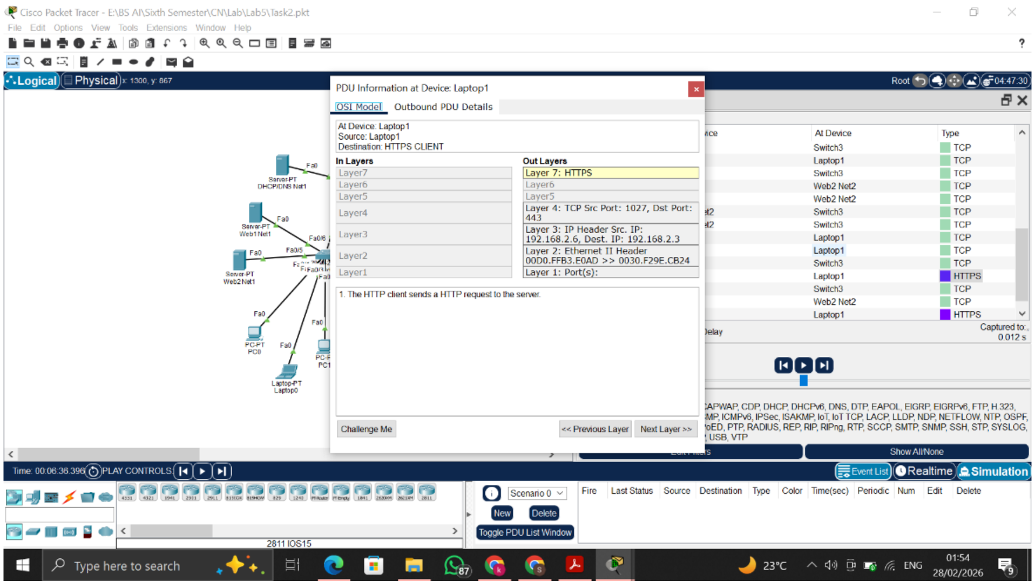

Laptop0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
Bandwidth
Duplex
MAC Address
000C.CF41.BA22
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.6
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Top
Type here to search
23°C
ENG
01:35
28/02/2026

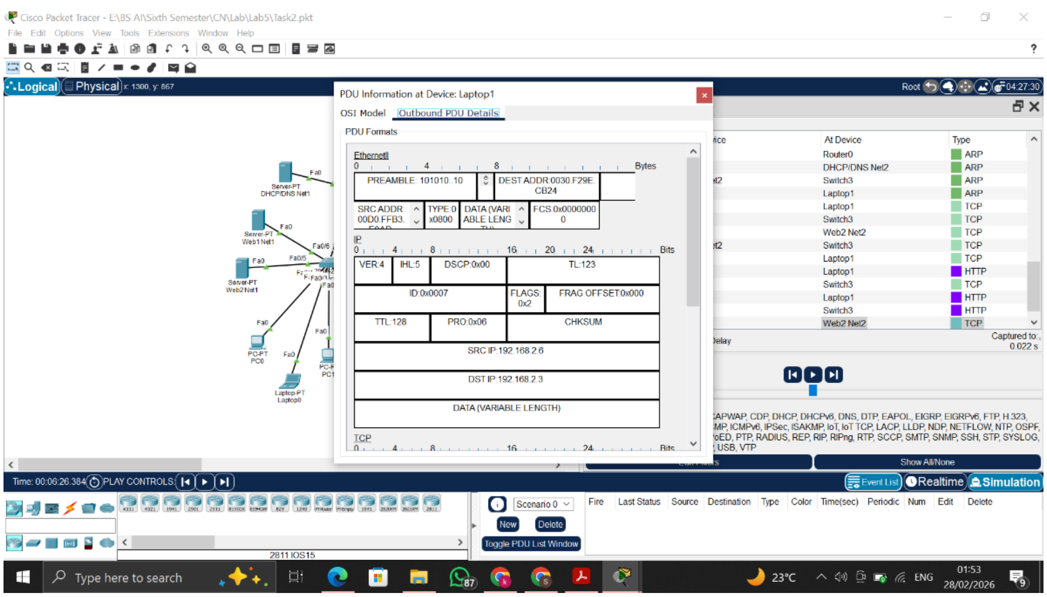

Laptop0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
Global Settings
Display Name
Laptop0
Interfaces
FastEthernet0
Gateway/DNS IPv4
DHCP
Static
Default Gateway
0.0.0.0
DNS Server
192.168.1.4
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top
Type here to search
23°C
ENG
01:35
28/02/2026

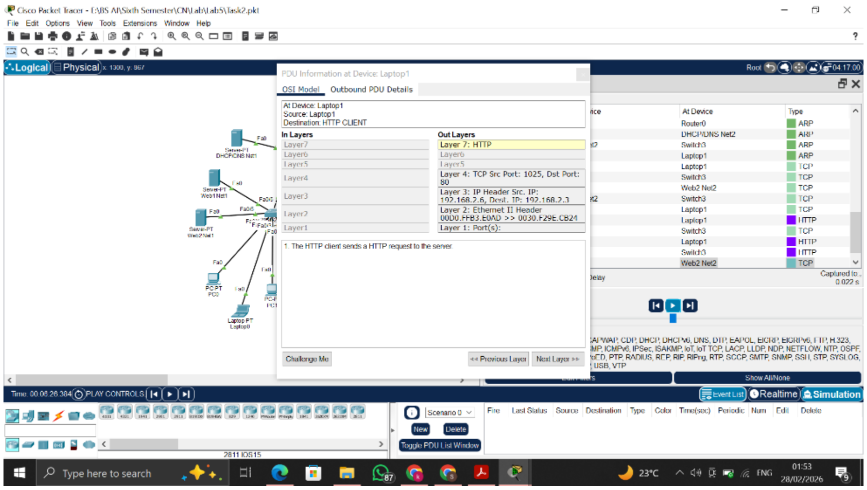

DHCP/DNS Net1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
192.168.1.4
Start IP Address
192
168
1
5
Subnet Mask
255
255
255
0
Maximum Number of Users
248
TFTP Server
0.0.0.0
WLC Address
0.0.0.0
Add
Save
Remove
Top
Type here to search
23°C
ENG
01:34
28/02/2026

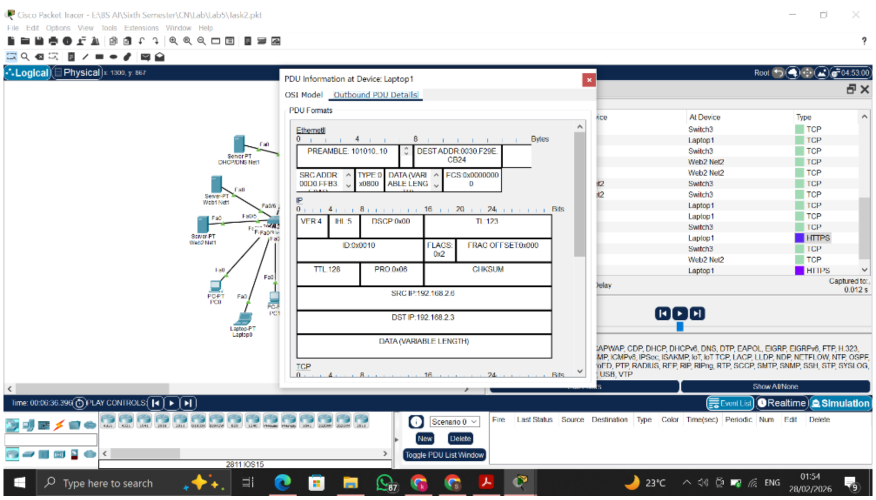
Lab Tasks 3:

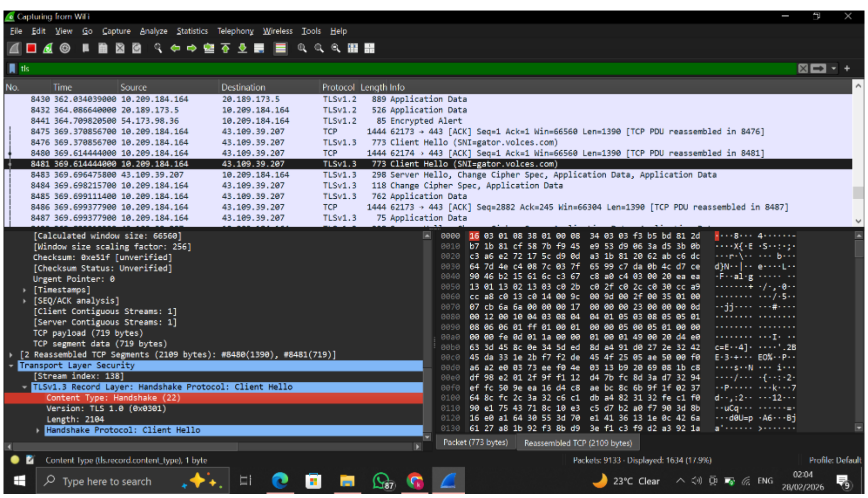
1. *Follow the above step for HTTPS. Take Snapshot of each Step, and Submit in docx file/pdf with one-line answer, what you understand here?*

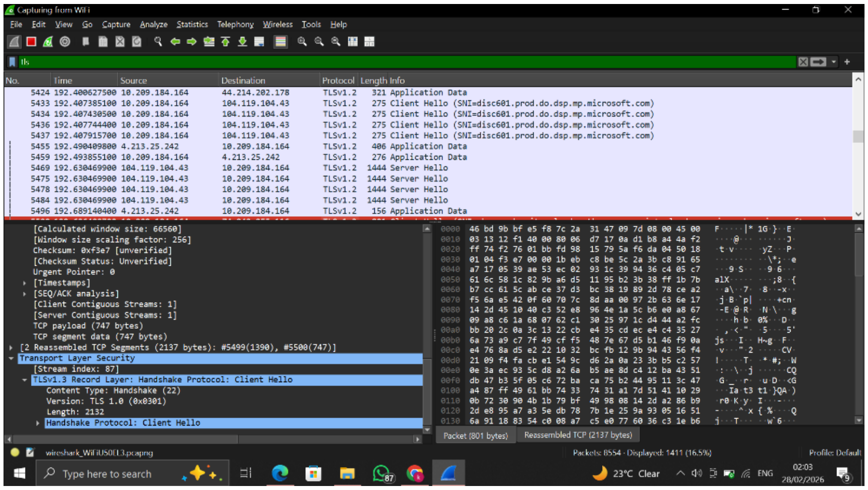


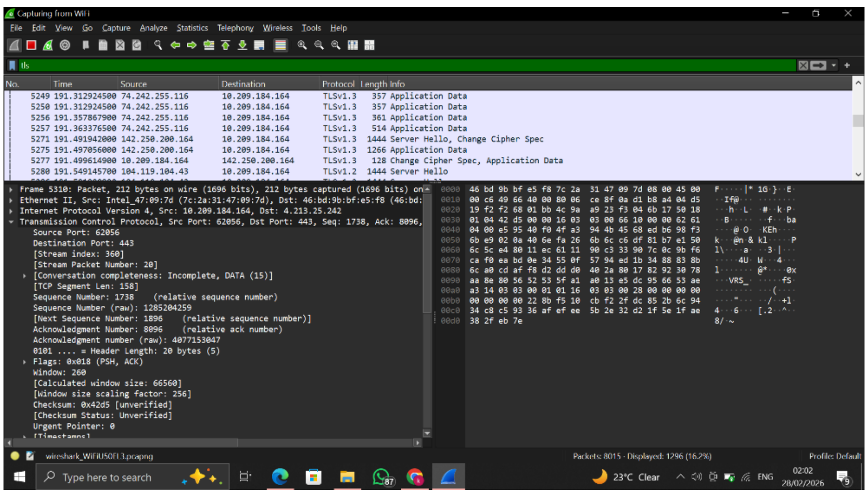


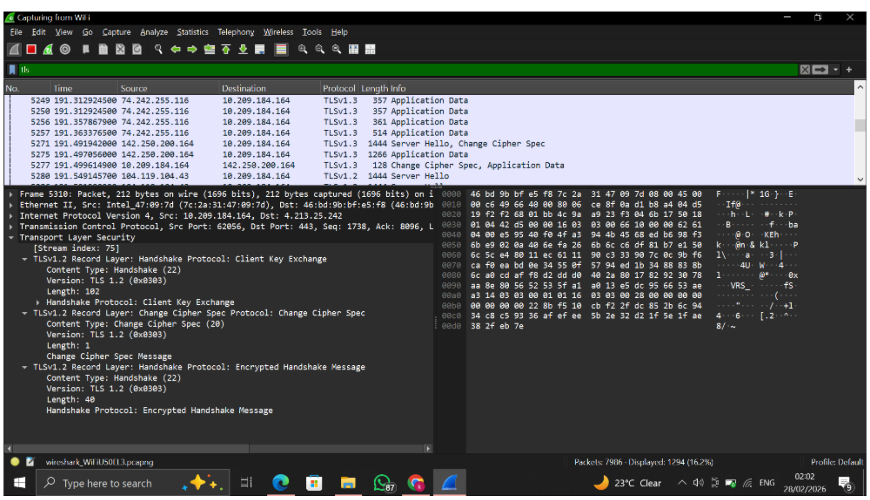

PDU Information at Device: Laptop1
OSI Model
Outbound PDU Details
At Device: Laptop1
Source: Laptop1
Destination: HTTP CLIENT
In Layers
Out Layers
Layer 7: HTTP
Layer 4: TCP Src Port: 1025, Dst Port: 80
Layer 3: IP Header Src. IP: 192.168.2.6, Dest. IP: 192.168.2.3
Layer 2: Ethernet II Header 00D0.FFB3.E0AD >> 0030.F29E.CB24
Layer 1: Port(s):
1. The HTTP client sends a HTTP request to the server.
Challenge Me
<< Previous Layer
Next Layer >>

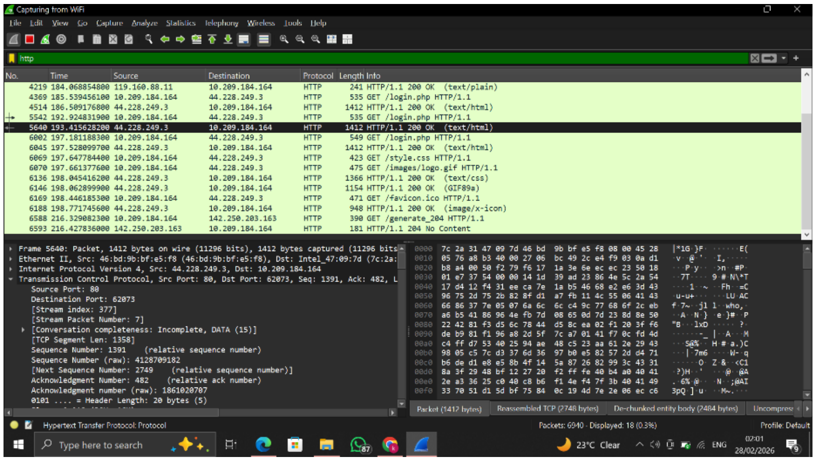

PDU Information at Device: Laptop1
OSI Model
Outbound PDU Details
PDU Formats
Ethernet II
PREAMBLE: 101010...10
DEST ADDR:0030.F29E.CB24
SRC ADDR: 00D0.FFB3
TYPE:0x0800
FCS:0x00000000
IP
VER:4
IHL:5
DSCP:0x00
TL:123
ID:0x0010
FLAGS: 0x2
FRAG OFFSET:0x000
TTL:128
PRO:0x06
CHKSUM
SRC IP:192.168.2.6
DST IP:192.168.2.3
DATA (VARIABLE LENGTH)
TCP

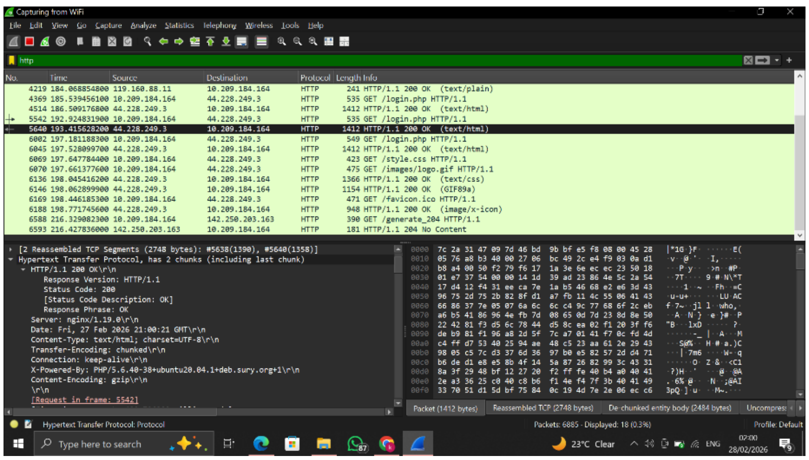

Capturing from WiFi
tls
8481 369.614444000 10.209.184.164 43.109.39.207 TLSv1.3 773 Client Hello (SNI=gator.volces.com)
8483 369.696475800 43.109.39.207 10.209.184.164 TLSv1.3 298 Server Hello, Change Cipher Spec, Application Data, Application Data
Transport Layer Security
TLSv1.3 Record Layer: Handshake Protocol: Client Hello
Content Type: Handshake (22)
Version: TLS 1.0 (0x0301)
Length: 2104
Handshake Protocol: Client Hello

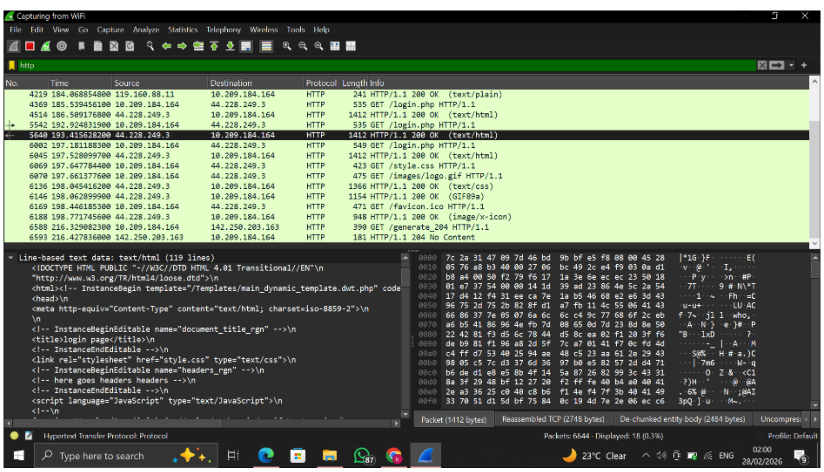

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tls
No. Time Source Destination Protocol Length Info
5424 192.400627500 10.209.184.164 44.214.202.178 TLSv1.2 321 Application Data
5433 192.407385100 10.209.184.164 104.119.104.43 TLSv1.2 275 Client Hello (SNI=disc601.prod.do.dsp.mp.microsoft.com)
5434 192.407430500 10.209.184.164 104.119.104.43 TLSv1.2 275 Client Hello (SNI=disc601.prod.do.dsp.mp.microsoft.com)
5436 192.407744400 10.209.184.164 104.119.104.43 TLSv1.2 275 Client Hello (SNI=disc601.prod.do.dsp.mp.microsoft.com)
5437 192.407915700 10.209.184.164 104.119.104.43 TLSv1.2 275 Client Hello (SNI=disc601.prod.do.dsp.mp.microsoft.com)
5455 192.490409800 4.213.25.242 10.209.184.164 TLSv1.2 406 Application Data
5459 192.493855100 10.209.184.164 4.213.25.242 TLSv1.2 276 Application Data
5469 192.630469900 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
5475 192.630469900 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
5478 192.630469900 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
5484 192.630469900 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
5496 192.689140400 4.213.25.242 10.209.184.164 TLSv1.2 156 Application Data
[Calculated window size: 66560]
[Window size scaling factor: 256]
Checksum: 0xf3e7 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (747 bytes)
TCP segment data (747 bytes)
[2 Reassembled TCP Segments (2137 bytes): #5499(1390), #5500(747)]
Transport Layer Security
[Stream index: 87]
TLSv1.3 Record Layer: Handshake Protocol: Client Hello
Content Type: Handshake (22)
Version: TLS 1.0 (0x0301)
Length: 2132
Handshake Protocol: Client Hello
Packet (801 bytes) Reassembled TCP (2137 bytes)
wireshark_WiFiUS0EL3.pcapng
Packets: 8554 · Displayed: 1411 (16.5%)
Profile: Default
Type here to search
23°C Clear
ENG
02:03
28/02/2026

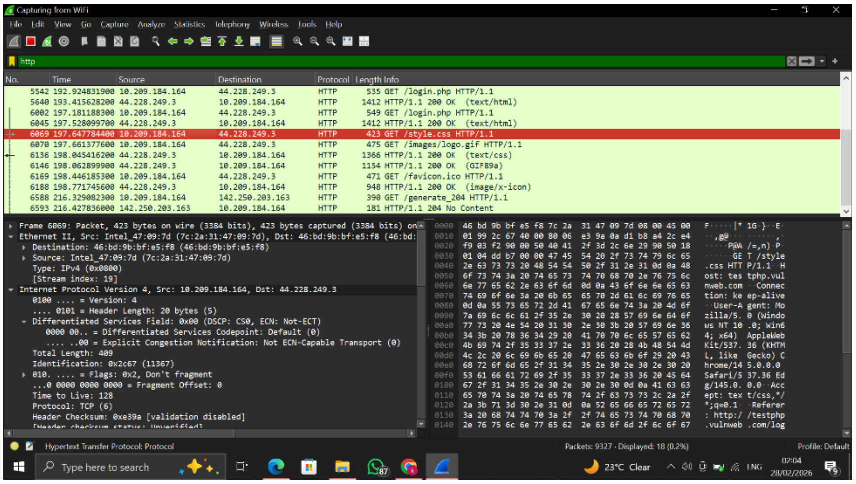

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tls
No. Time Source Destination Protocol Length Info
5249 191.312924500 74.242.255.116 10.209.184.164 TLSv1.3 357 Application Data
5250 191.312924500 74.242.255.116 10.209.184.164 TLSv1.3 357 Application Data
5256 191.357867900 74.242.255.116 10.209.184.164 TLSv1.3 361 Application Data
5257 191.363376500 74.242.255.116 10.209.184.164 TLSv1.3 514 Application Data
5271 191.491942000 142.250.200.164 10.209.184.164 TLSv1.3 1444 Server Hello, Change Cipher Spec
5275 191.497056000 142.250.200.164 10.209.184.164 TLSv1.3 1266 Application Data
5277 191.499614900 10.209.184.164 142.250.200.164 TLSv1.3 128 Change Cipher Spec, Application Data
5280 191.549145700 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
Frame 5310: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on
Ethernet II, Src: Intel_47:09:7d (7c:2a:31:47:09:7d), Dst: 46:bd:9b:bf:e5:f8 (46:bd:
Internet Protocol Version 4, Src: 10.209.184.164, Dst: 4.213.25.242
Transmission Control Protocol, Src Port: 62056, Dst Port: 443, Seq: 1738, Ack: 8096,
Source Port: 62056
Destination Port: 443
[Stream index: 360]
[Stream Packet Number: 20]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 158]
Sequence Number: 1738 (relative sequence number)
Sequence Number (raw): 1285204259
[Next Sequence Number: 1896 (relative sequence number)]
Acknowledgment Number: 8096 (relative ack number)
Acknowledgment number (raw): 4077153047
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 260
[Calculated window size: 66560]
[Window size scaling factor: 256]
Checksum: 0x42d5 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
wireshark_WiFiUS0EL3.pcapng
Packets: 8015 · Displayed: 1296 (16.2%)
Profile: Default
Type here to search
23°C Clear
ENG
02:02
28/02/2026


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tls
No. Time Source Destination Protocol Length Info
5249 191.312924500 74.242.255.116 10.209.184.164 TLSv1.3 357 Application Data
5250 191.312924500 74.242.255.116 10.209.184.164 TLSv1.3 357 Application Data
5256 191.357867900 74.242.255.116 10.209.184.164 TLSv1.3 361 Application Data
5257 191.363376500 74.242.255.116 10.209.184.164 TLSv1.3 514 Application Data
5271 191.491942000 142.250.200.164 10.209.184.164 TLSv1.3 1444 Server Hello, Change Cipher Spec
5275 191.497056000 142.250.200.164 10.209.184.164 TLSv1.3 1266 Application Data
5277 191.499614900 10.209.184.164 142.250.200.164 TLSv1.3 128 Change Cipher Spec, Application Data
5280 191.549145700 104.119.104.43 10.209.184.164 TLSv1.2 1444 Server Hello
Frame 5310: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on i
Ethernet II, Src: Intel_47:09:7d (7c:2a:31:47:09:7d), Dst: 46:bd:9b:bf:e5:f8 (46:bd:9b
Internet Protocol Version 4, Src: 10.209.184.164, Dst: 4.213.25.242
Transmission Control Protocol, Src Port: 62056, Dst Port: 443, Seq: 1738, Ack: 8096, L
Transport Layer Security
[Stream index: 75]
TLSv1.2 Record Layer: Handshake Protocol: Client Key Exchange
Content Type: Handshake (22)
Version: TLS 1.2 (0x0303)
Length: 102
Handshake Protocol: Client Key Exchange
TLSv1.2 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
Content Type: Change Cipher Spec (20)
Version: TLS 1.2 (0x0303)
Length: 1
Change Cipher Spec Message
TLSv1.2 Record Layer: Handshake Protocol: Encrypted Handshake Message
Content Type: Handshake (22)
Version: TLS 1.2 (0x0303)
Length: 40
Handshake Protocol: Encrypted Handshake Message
wireshark_WiFiUS0EL3.pcapng
Packets: 7986 · Displayed: 1294 (16.2%)
Profile: Default
Type here to search
23°C Clear
ENG
02:02
28/02/2026


Capturing from WiFi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Source Destination Protocol Length Info
4219 184.068854800 119.160.88.11 10.209.184.164 HTTP 241 HTTP/1.1 200 OK (text/plain)
4369 185.539456100 10.209.184.164 44.228.249.3 HTTP 535 GET /login.php HTTP/1.1
4514 186.509176800 44.228.249.3 10.209.184.164 HTTP 1412 HTTP/1.1 200 OK (text/html)
5542 192.924831900 10.209.184.164 44.228.249.3 HTTP 535 GET /login.php HTTP/1.1
5640 193.415628200 44.228.249.3 10.209.184.164 HTTP 1412 HTTP/1.1 200 OK (text/html)
6002 197.181188300 10.209.184.164 44.228.249.3 HTTP 549 GET /login.php HTTP/1.1
6045 197.528099700 44.228.249.3 10.209.184.164 HTTP 1412 HTTP/1.1 200 OK (text/html)
6069 197.647784400 10.209.184.164 44.228.249.3 HTTP 423 GET /style.css HTTP/1.1
6070 197.661377600 10.209.184.164 44.228.249.3 HTTP 475 GET /images/logo.gif HTTP/1.1
6136 198.045416200 44.228.249.3 10.209.184.164 HTTP 1366 HTTP/1.1 200 OK (text/css)
6146 198.062899900 44.228.249.3 10.209.184.164 HTTP 1154 HTTP/1.1 200 OK (GIF89a)
6169 198.446185300 10.209.184.164 44.228.249.3 HTTP 471 GET /favicon.ico HTTP/1.1
6188 198.771745600 44.228.249.3 10.209.184.164 HTTP 948 HTTP/1.1 200 OK (image/x-icon)
6588 216.329082300 10.209.184.164 142.250.203.163 HTTP 390 GET /generate_204 HTTP/1.1
6593 216.427836000 142.250.203.163 10.209.184.164 HTTP 181 HTTP/1.1 204 No Content
Frame 5640: Packet, 1412 bytes on wire (11296 bits), 1412 bytes captured (11296 bits
Internet Protocol Version 4, Src: 44.228.249.3, Dst: 10.209.184.164
Source Port: 80
Destination Port: 62073
[Stream index: 377]
[Stream Packet Number: 7]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 1358]
Sequence Number: 1391 (relative sequence number)
Sequence Number (raw): 4128709182
[Next Sequence Number: 2749 (relative sequence number)]
Acknowledgment Number: 482 (relative ack number)
Acknowledgment number (raw): 1861020707
0101 .... = Header Length: 20 bytes (5)
Packet (1412 bytes) Reassembled TCP (2748 bytes) De-chunked entity body (2484 bytes)
Hypertext Transfer Protocol: Protocol
Packets: 6940 · Displayed: 18 (0.3%)
Profile: Default
Type here to search
23°C Clear
ENG
07:01
28/02/2026


Capturing from WiFi
[2 Reassembled TCP Segments (2748 bytes): #5638(1390), #5640(1358)]
Hypertext Transfer Protocol, has 2 chunks (including last chunk)
HTTP/1.1 200 OK\r\n
Response Version: HTTP/1.1
Status Code: 200
[Status Code Description: OK]
Response Phrase: OK
Server: nginx/1.19.0\r\n
Date: Fri, 27 Feb 2026 21:00:21 GMT\r\n
Content-Type: text/html; charset=UTF-8\r\n
Transfer-Encoding: chunked\r\n
Connection: keep-alive\r\n
X-Powered-By: PHP/5.6.40-38+ubuntu20.04.1+deb.sury.org+1\r\n
Content-Encoding: gzip\r\n
\r\n
[Request in frame: 5542]
Hypertext Transfer Protocol: Protocol
Packets: 6885 · Displayed: 18 (0.3%)
Profile: Default
07:00
28/02/2026


Capturing from WiFi
Line-based text data: text/html (119 lines)
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"\n
"http://www.w3.org/TR/html4/loose.dtd">\n
<html><!-- InstanceBegin template="/Templates/main_dynamic_template.dwt.php" code
<head>\n
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-2">\n
\n
<!-- InstanceBeginEditable name="document_title_rgn" -->\n
<title>login page</title>\n
<!-- InstanceEndEditable -->\n
<link rel="stylesheet" href="style.css" type="text/css">\n
<!-- InstanceBeginEditable name="headers_rgn" -->\n
<!-- here goes headers headers -->\n
<!-- InstanceEndEditable -->\n
<script language="JavaScript" type="text/JavaScript">\n
<!--\n
Hypertext Transfer Protocol: Protocol
Packets: 6644 · Displayed: 18 (0.3%)
Profile: Default
07:00
28/02/2026



HTTPS uses TLS encryption to secure the data, so the packet content is not readable in Wireshark.

2. *Observe the difference between HTTP and HTTPS and answer in one line with proper snapshots?*

HTTP sends data in plain text format on port 80, so the data is readable in Wireshark. While HTTPS sends encrypted data using the TLS library on port 443, the data is not in a readable format